Department of Artificial Intelligence,SVNIT
Subject- Machine Learning(AI301)

Background:
You are given a dataset containing 79 explanatory variables describing (almost) every aspect of residential homes in Ames, Iowa. The target variable is SalePrice. Your task is to perform **data preprocessing** and **exploratory analysis** to prepare the dataset for predictive modeling.

Part A: Data Loading & Basic Inspection

- Load the training dataset into a Pandas DataFrame. Display the first 10 rows.
- How many rows and columns does the dataset have?
- Identify the **numerical** and **categorical** columns. How many are there of each type?
- Check the datatypes of all columns.Are there any incorrect datatypes that need conversion?

Part B: Univariate Analysis

- For the **numerical columns**:
    1. Plot histograms for atleast 5 key features (LotArea,GrLivArea,SalePrice, YearBuilt, OverallQual).
    2. Identify whether the distributions are skewed or approximately normal.
- For the **categorical columns**:
    1. Plot bar charts for at least 3 variables (Neighborhood, HouseStyle, BldgType).
    2. Which categories are most common and which are rare?
- Show value counts for the top 10 most frequent categories across all categorical features.
- Identify any **imbalanced categories** where one category dominates.

Part C: Bivariate Analysis

- Plot a scatter plot between GrLivArea and SalePrice. What trend do you observe?
- Plot a boxplot of SalePrice across different OverallQual values.
- Create a correlation heatmap for the top 10 features most correlated with SalePrice.
- Compare average SalePrice between two neighborhoods of your choice. Is there a significant difference?
- Compare SalePrice between newer homes (YearBuilt >= 2000) and older homes. Use both:
    1. Boxplot
    2. T-test for mean difference

Part D: Missing Value Treatment

- Find the total number of missing values in each column. List the top 5 columns with the most missing values.
- For numerical columns, impute missing values using median.
- For categorical columns, impute missing values using mode or a placeholder like "Missing".
- For LotFrontage specifically, impute using the median value grouped by Neighborhood.
- After imputation, verify there are no missing values left.

Part E: Outlier Detection & Treatment

- Use the IQR method to detect outliers in LotArea and GrLivArea.
- How many outliers are detected in each feature?
- Remove or cap extreme outliers and explain why you chose that method.
- Replot the scatter plot from Q7 after removing/capping outliers. Has the relationship become clearer?

Part F: Feature Scaling

- Standardize numerical features using **StandardScaler** or **MinMaxScaler**.
- Save the cleaned and processed dataset as house_prices_clean.csv.

Dataset:
Kaggle – House Prices: Advanced Regression Techniques

Part G : K-Nearest Neighbour Classification

- Implement a **K-Nearest Neighbour (KNN) classification algorithm in Python** to classify samples from the **Iris dataset** into its three flower species: *Setosa, Versicolor,* and *Virginica*.
- The dataset contains four features: **sepal length, sepal width, petal length, and petal width**. Divide the dataset into training and testing sets, normalize the features, and classify the test samples using KNN based on distance and majority voting.
- Investigate the effect of different values of **K** on classification performance and compare different distance measures such as **Euclidean and Manhattan distance**. Evaluate the classifier using appropriate performance metrics and determine the value of K that provides the best classification performance.